POLITECNICO DI MILANO - DIPARTIMENTO DI FISICA

[redacted]	DATA/DATE 10/02/25
ESAME DI/EXAMINATION OF	SPAZIO RISERVATO/RESERVED SPACE 1) errori [ore $\Delta\varphi$] 12 2) OK def. 12. Mo dim. cancellat.
DOCENTE/PROFESSOR	

es 2)

a) Radiance (L) is defined as the [redacted] element of radiant flux [redacted] [W] which travels through a surface element $dA_\perp$ and a certain solid angle $d\Omega$.
It's unit of measure is $W/m^2 \cdot sr$ and can be calculated as

$$L = \frac{d\Phi}{\underbrace{dA_\perp}\, d\Omega} = \frac{d\Phi}{dA \cos\theta \, d\Omega}$$

↳ this means that the ~~[redacted]~~ area considered is the one normal to the radiant flux

It applies only in lossless mediums.

b) We have that the condition required for the conservation of radiance is the fact that the radiant field must travel through a lossless medium

This happens because if the medium isn't lossless we would have a loss of power per unit of surface through the medium. This will result in a decrease of the radiance.

c) If we define the radiant flux coming from a surface as

$$d\Phi_0 = \pi L \cos\theta_0 \, dA_0 \, d\Omega_0$$

non è necessario che la sorgente sia lambertiana

~~[redacted] and the flux going to another surface is~~

while the radiant flux hitting on another surface is defined as

$$d\Phi_1 = \pi L \cos\theta_1 \, dA_1 \, d\Omega_1$$

~~We have to use the Abbe condition and its derivative~~

We get that $dA_0 \propto dx_0 dy_0$
$dA_1 \propto dx_1 dy_1$

By the use of Abbe's condition and its derivative along y axis $n_0 dx_0 \sin\theta_0 = n_1 dx_1 \sin\theta_1$
we compare $d\Phi_0 = d\Phi_1$ we get that

$$L_0 / n_0^2 = L_1 / n_1^2$$

so if [redacted] $n_0 = n_1 =$ refractive index we get that $L_1 = L_0$

es 1)

a) $NEP = \frac{i_{noise}^{RMS}}{R_\lambda} = \frac{hc}{\lambda \eta e} \cdot \boxed{i_{noise}^{RMS}} = \frac{1.57 \cdot 10^{-9}}{4}$ W

$i_{noise}^{RMS} = \sqrt{\left(2eG^2F_A I_d + \frac{4k_BT}{R_L}\right)\Delta f}$

$= 8.139 \cdot 10^{-9}$ A OK

b) modulation depth $= 1$ $\Rightarrow$ $\Phi_0^{RMS} = \frac{\Phi_0}{\sqrt{2}}$

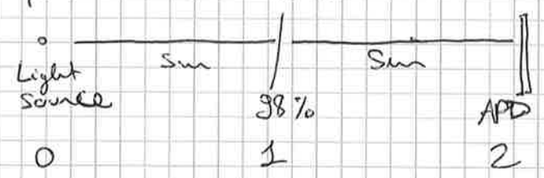
$\Phi_0 = P(1 + \sin(2\pi f t))$

where $f = 5$ MHz

$d_{tot} = 2H = 10$ m

However:

Equivalent circuit:



$E_1 = \frac{d\Phi}{dA} = \frac{\Phi_0^{RMS}}{\pi H^2}$

$M_1 = 0.98 \cdot \frac{\Phi_0}{\pi H^2} \Rightarrow L_1 = \frac{M_1}{M}$

$L_1 = \frac{0.98 \cdot \Phi_0}{\pi^2 H^2}$? $\Rightarrow E = L_1 \cdot \frac{}{4\pi H^2}$

$\Phi_{av} = \pi L_1 \cdot \underbrace{\pi \left(\frac{D}{2}\right)^2}_{Area}$ HC $\pi^2 L_1 \left(\frac{D}{2}\right)^2 = \frac{\pi^2}{4} L_1 D^2 = \frac{\pi^2}{4} D^2 \cdot 0.98 \frac{\Phi_0}{\pi^2 H^2} = \frac{0.98 D^2 \Phi_0^{RMS}}{4H^2}$

$\approx 7 \cdot 10^{-9}$ W

c) $\langle i_s \rangle = R_\lambda \cdot \Phi_{av}^{RMS}$ where $R_\lambda = \eta \frac{e\lambda}{hc} = 0.517 \frac{A}{W}$

$= 3.6 \cdot 10^{-9}$ A

d) $i_{tot} = i_{signal} + i_{noise}$

$= G\langle i_s \rangle + i_{noise}$

where $i_{noise} = \sqrt{\left(2eG^2(\langle i_s \rangle + I_d)F_A + \frac{4k_BT}{R_L}\right)\Delta f}$

$i_{signal} = G\langle i_s \rangle$

e) $SNR = \frac{i_{signal}}{i_{noise}} = \frac{\langle i_s \rangle}{\sqrt{\left(2eF_A(I_d + \langle i_s \rangle) + \frac{4k_BT}{G^2R_L}\right)\Delta f}}$

d) $\Delta\varphi = \frac{2\pi}{\lambda} 2H$ where $\lambda = \frac{c}{f} = \frac{3 \cdot 10^8}{5\,MHz} = 60$ m

$= \frac{2\pi}{60\,m} 5\,m = 2\frac{\pi}{6}$ Modulation in intensity OK